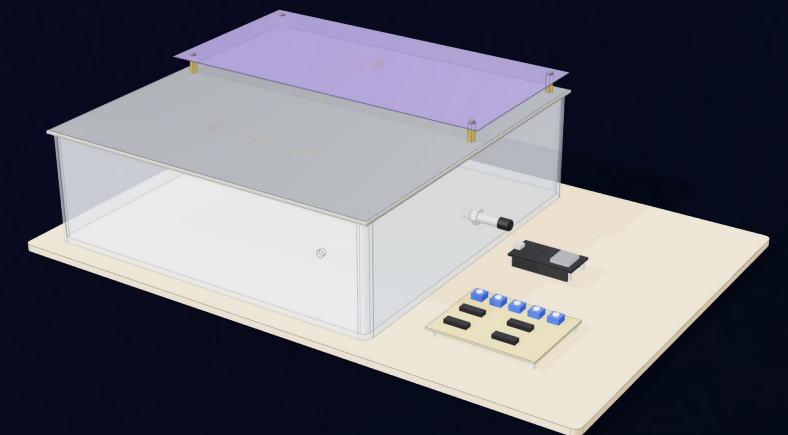


SEDHACKS'26

HULLSENSE

Where did it hit? Did it breach? How long does the crew have? **Answered in seconds, for ₹1,479.**

- Problem Statement Title: **Real-time strike localisation and breach sizing for crewed-module walls**
- Track: **03 • Space Instrumentation**
- Category: **Hardware**
- Team Lead Name: **YOGEESHWARAN C**
- Team Name: **BYTEMELATER**



PROPOSED SOLUTION

THE SOLUTION

HULLSENSE is a ₹1,479 sensor skin for a crewed-module wall. When something strikes it, the system tells the crew **where it hit, whether the hull was breached, how big the hole is and how long they have**, within seconds.

Locate the strike

Four corner sensors time the ripple a strike sends through the wall. The arrival-time differences give the (x, y) point.

Shield vs hull

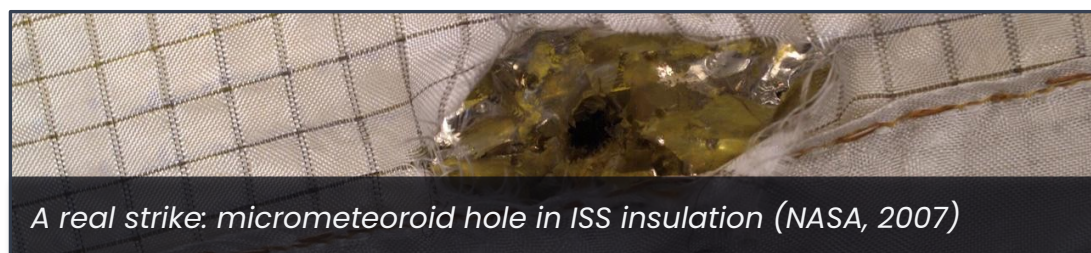
A fifth sensor on the outer bumper tells a harmless shield hit apart from a strike on the pressure wall.

Size the breach

If box pressure falls after a wall strike, the decay curve gives the hole diameter and the minutes left.

Alert in seconds

The dashboard shows the verdict, the location and the pressure fit as soon as the strike lands.



A real strike: micrometeoroid hole in ISS insulation (NASA, 2007)

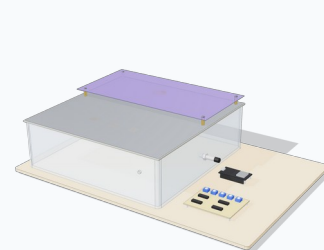
PROTOTYPE / SYSTEM DESIGN

Sensor node



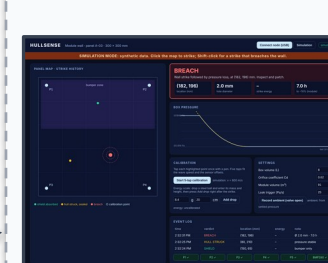
- 5 piezo discs, ₹20 each
- BMP280 pressure sensor
- LM324 band-pass filters
- LM339 comparators
- ESP32 capture timer

Test wall



CAD model: 300 mm aluminium wall and bumper on a sealed 8 L box

Dashboard

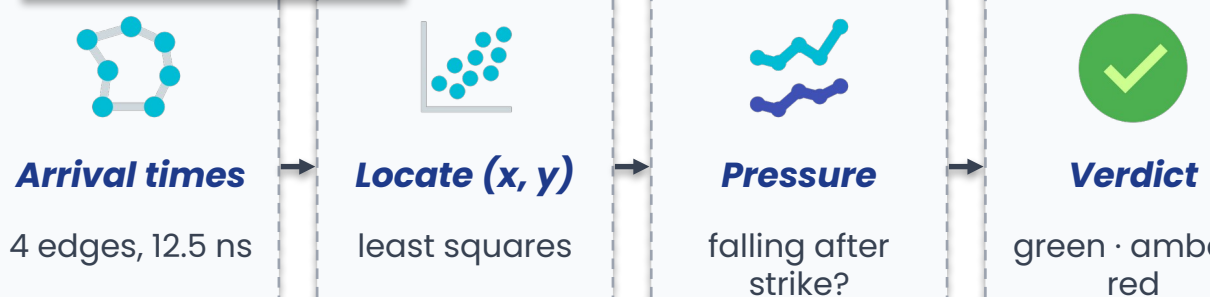


Working web app: verdict, map and pressure fit (simulation shown)

Data flow



Decision pipeline



Sense → Locate → Decide → Alert

NOVELTY



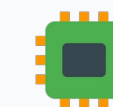
Strike + breach in one system

NASA's DIDS only detects. Honeywell only sizes holes.



Two-layer triage

Separates shield hits from pressure-wall hits.



12.5 ns timing, no fast ADC

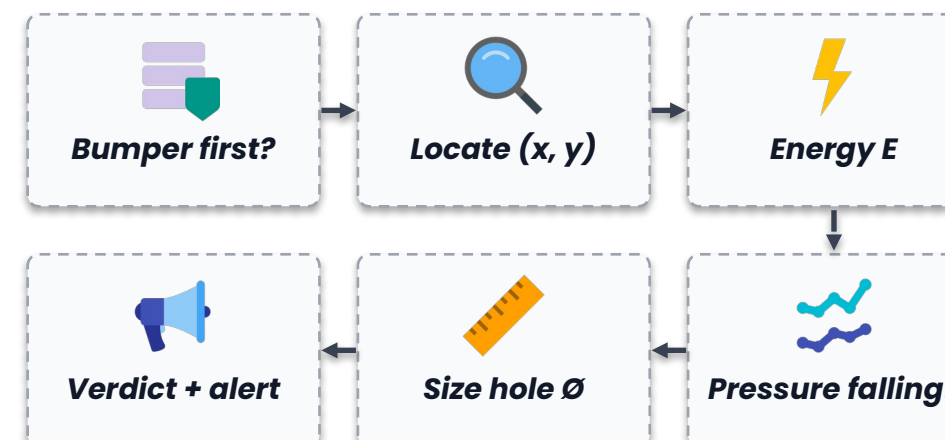
Hardware edge capture replaces a 1 MHz DAQ.



Self-calibrating

Five taps fit wave speed and sensor offsets.

UNIQUE DECISION LOGIC



Physics-first: wave speed and hole size come from equations, calibrated on the bench.

Fail-safe: a sensor that goes silent raises a fault, never an "all clear".

PHYSICS + SIGNAL

Where it hit: arrival times

$$t_i = t_0 + |p - s_i| / v$$

Four equations, three unknowns: the solver returns (x, y) plus a confidence.

How big: orifice blow-down

$$dP/dt = -(P/V) \cdot C_d \cdot A \cdot \sqrt{(2\Delta P/\rho)}$$

Fitting the pressure curve gives the hole area, then its diameter.

How long: choked flow to vacuum

$$\tau = V / (C_d \cdot A \cdot c^*), \quad c^* \approx 199 \text{ m/s}$$

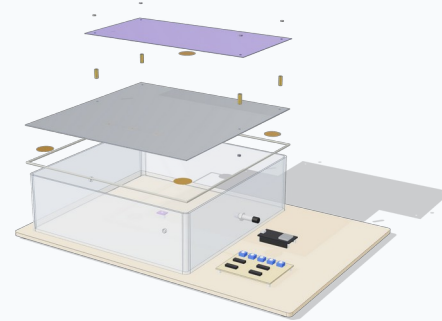
Scales the hole to a real module: minutes to a set pressure.



Timing tells us where it hit.
Pressure tells us how bad it is.

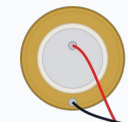
SYSTEM ARCHITECTURE & FLOW

Structure layer



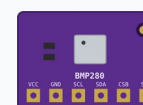
CAD, exploded: bumper on 15 mm standoffs over a 1 mm wall that lids a sealed 8 L box

Sensing layer



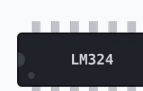
5 piezo discs

4 on the wall corners, 1 on the bumper



BMP280

pressure sensor inside the sealed box



LM324 + LM339

band-pass filter, then threshold

Processing layer



Capture: 12.5 ns edge stamps

I2C: pressure at ~100 Hz

USB: JSON events to laptop

Decision layer

Shield absorbed

Hull struck, sealed

Breach: 0+ minutes

Data pipeline



Ripple



Band-pass



Comparator



Capture



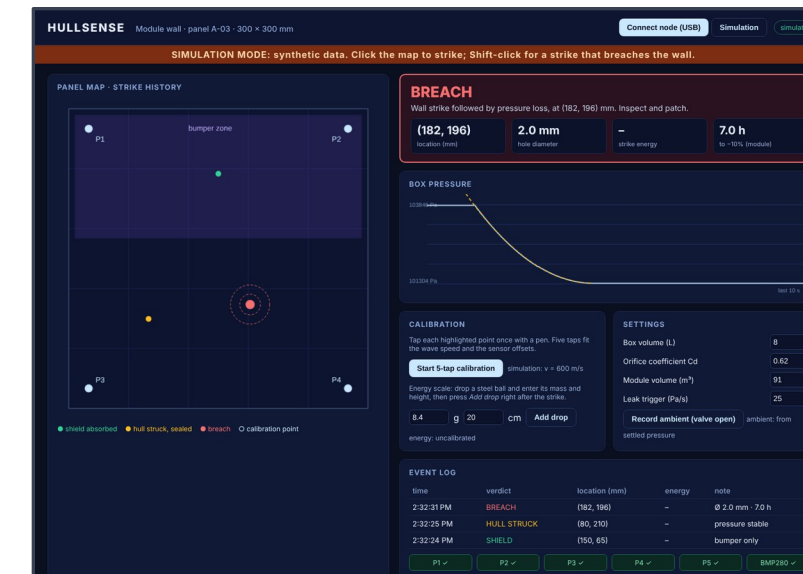
Solver



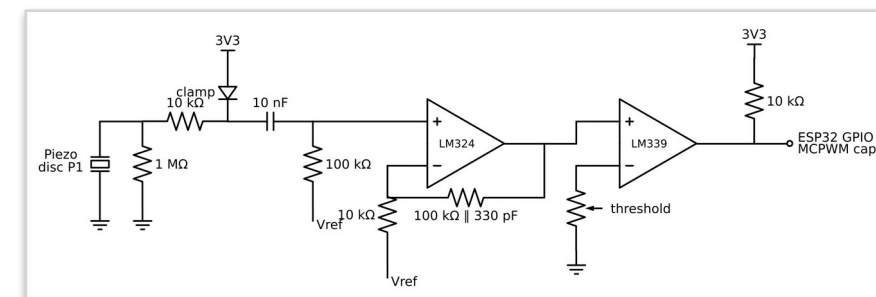
Alert

WORKING PROTOTYPE

Dashboard (real app, simulation mode)



Node circuit, one of five channels



Firmware compiles for ESP32 (317 KB). Code on GitHub.

METHODOLOGY & IMPLEMENTATION FLOW



Build

wall, bumper and sealed box



Calibrate

5 taps and 3 ball drops



Capture

12.5 ns edges, 100 Hz pressure



Locate

least-squares (x, y)



Decide

triage and leak fit



Alert

verdict on the dashboard

TECHNOLOGIES USED

Hardware

ESP32 · piezo · BMP280 · LM324 · LM339

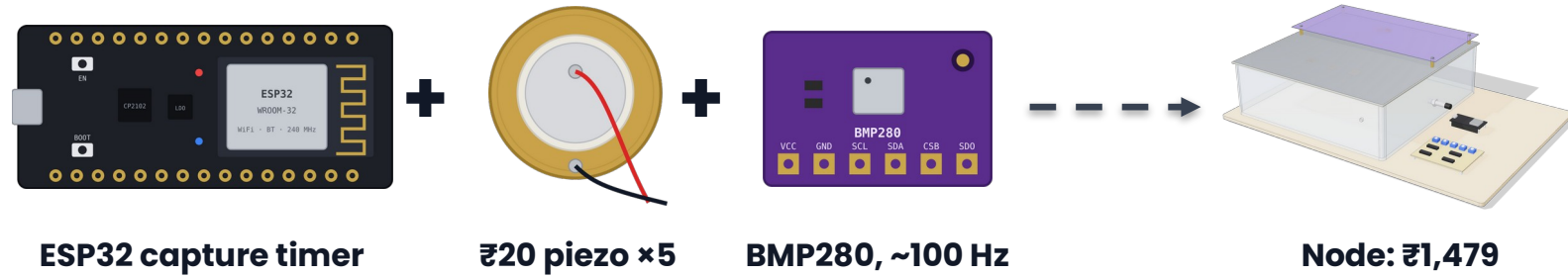
Firmware

Arduino-ESP32 3.x (ESP-IDF MCPWM driver)

Software

JavaScript Web Serial · Python · NumPy

TECHNICAL FEASIBILITY



12.5 ns timing

Ripples cross the plate at hundreds of m/s, so this resolves millimetres.



1, 2 and 3 mm holes

drain the box in 11, 2.7 and 1.2 s: long enough to fit reliably.



≤ 2.5 cm target

median location error, tested on a 25-point grid before the event.

OPERATIONAL FEASIBILITY

- **Fits under an existing wall panel:** it bolts on and needs no change to the hull.
- **Works offline:** ESP32 to laptop over USB; no Wi-Fi or internet in the loop.

ECONOMICAL FEASIBILITY

- **₹1,479 per node.** ESP32, piezo and BMP280 prices checked online; the rest are estimates.
- **₹14,790** for a 10-panel test module [est].

OUR PROMISES

- **Locate** the strike in seconds, **size** the breach, **₹1,479** per wall panel.

VIABILITY



Social: Gives the crew a clear time budget instead of a guess.



Market: Structural health monitoring is US\$2.7B (2025), heading to US\$6.7B by 2034.



Adoption: Off-the-shelf parts that can be bought and built in 48 h from Ritchie Street.



Policy fit: Supports India's path from Gaganyaan to the Bharatiya Antariksh Station (first module targeted 2028).



Scalability: One node per wall panel, running the same firmware on every panel.



Financial: The whole node costs 1/14 of one industrial acoustic-emission sensor (₹21,449).

SWOT ANALYSIS

STRENGTH

Locates strikes with ₹20 sensors; the whole node costs ₹1,479.

WEAKNESS

Tested with lab strikes so far; no hypervelocity test yet.

SWOT

THREAT

Space qualification (vacuum, radiation, thermal) is slow and costly.

OPPORTUNITY

India's space economy: US\$8.4B today, US\$44B targeted by 2033.

IMPACT AT A GLANCE



SAFETY IMPACT

Hidden damage becomes visible within seconds.



SCALE IMPACT

The same node fits every wall panel.



MISSION IMPACT

The crew gets minutes left, not a guess.

ECONOMIC BENEFIT



At ₹1,479 per panel, catching a breach early costs far less than losing a module or a mission.

SAFETY BENEFIT



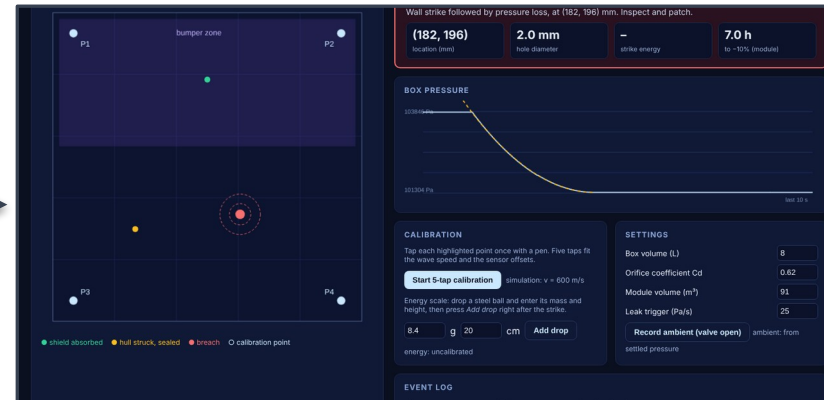
Protects the crews of India's first space station, where help is hours away.

STAKEHOLDER JOURNEY



1 • Debris strikes a module wall

Damage like this can go unnoticed for months.



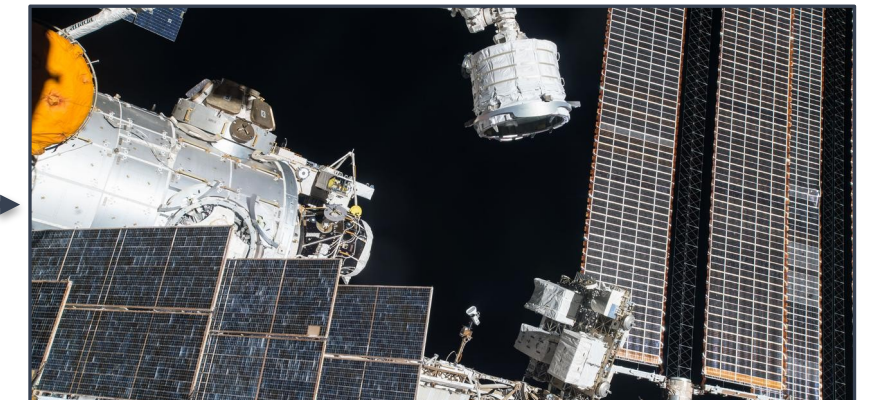
2 • HULLSENSE locates and sizes it

Within seconds, on the crew dashboard.



3 • Crew patches or closes the hatch

With a time budget, not a guess.



4 • The station keeps flying

Every strike is logged to improve future shields.

A silent breach becomes a located, sized alert, with minutes to act.

WIDER IMPACT



Students & startups

Affordable structural monitoring for small satellites.



Engineers

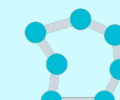
Per-panel impact data for maintenance.



National

Supports India's human-spaceflight goals.

TECHNICAL BENEFIT



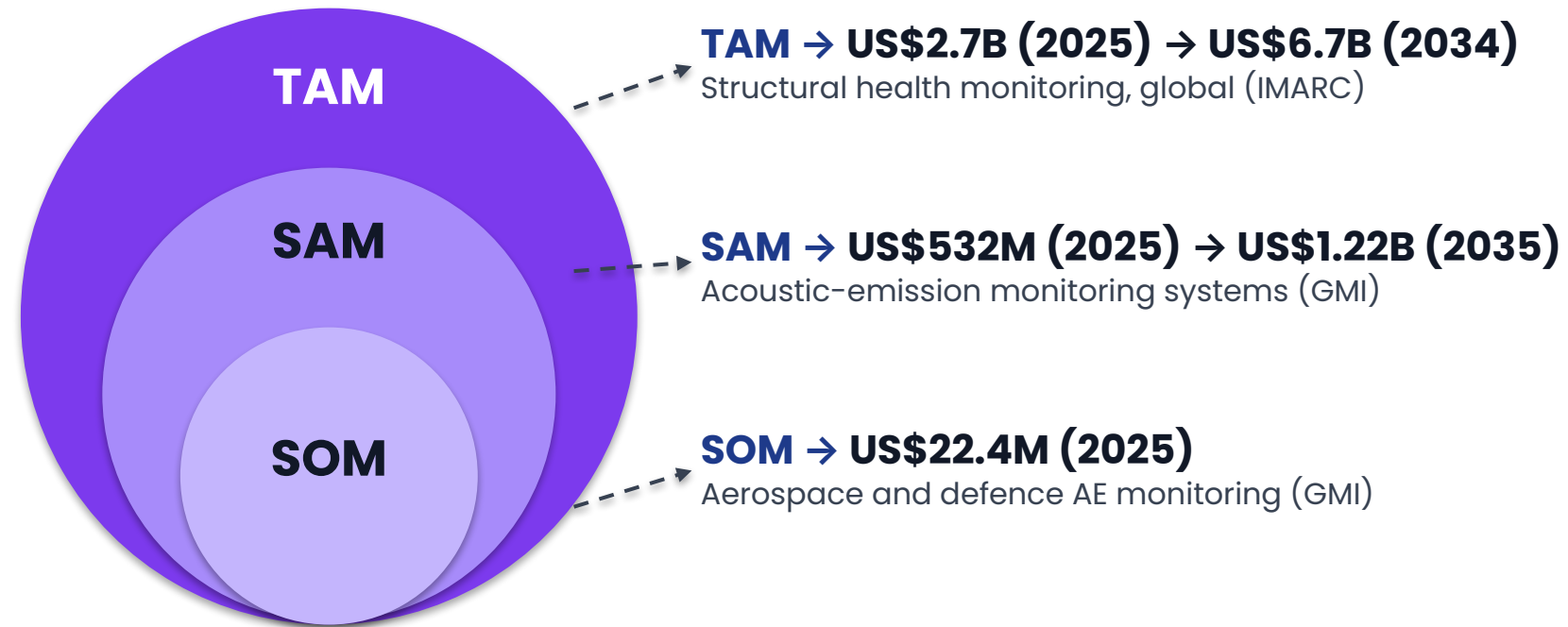
Hears strikes from debris too small for ground radar to track.

RESILIENCE



A sensor that goes silent raises a fault: silence is never "all clear".

MARKET ANALYSIS



INDIA'S SPACE ECONOMY

US\$8.4B → US\$44B

India's space economy today, and the target for 2033 stated by the IN-SPACe chairman. Crewed stations are part of that plan.



DEPLOYMENT ECONOMICS

- Hardware per node: ₹1,479
 - 10-panel test module: ₹14,790 [est]
 - One industrial AE sensor: ₹21,449+
- A full node costs less than one sensor.*

OUR WORK



Source code: firmware, dashboard and tests github.com/yogeesh2007-bit/hullsense



Documentation: build brief, wiring and BOM [raw.githubusercontent.com/.../hullsense/main/docs](https://raw.githubusercontent.com/yogeesh2007-bit/hullsense/main/docs)



Live dashboard: runs in any browser [raw.githubusercontent.com/.../docs/dashboard](https://raw.githubusercontent.com/yogeesh2007-bit/hullsense/main/docs/dashboard)



CAD model: 3D viewer, STEP, STL, drawing [raw.githubusercontent.com/.../docs/cad](https://raw.githubusercontent.com/yogeesh2007-bit/hullsense/main/docs/cad)

PRIOR ART & REFERENCES

	Locates	Shield/hull	Breach	Hole Ø	Cost
NASA DIDS (ISS)	Approx.	Offline	—	—	Flight HW
Honeywell US 6,885,967	Module	—	✓	✓	Patent
TapTDOA (hobby)	✓	—	—	—	Low
HULLSENSE	✓	✓	✓	✓	₹1,479

1. Wells & Madaras, NASA (2017): DIDS on BEAM
2. Corsaro et al., NASA HDU: 147/147 within 8 cm
3. Rachlin, Honeywell, US 6,885,967 (2005)
4. IMARC (2025): SHM market
5. GMI (2025): AE monitoring market

6. WION (2023): IN-SPACe US\$44B target
7. TapTDOA (Hackaday.io)
8. Espressif ESP-IDF: MCPWM capture
9. Photos: NASA (public domain)

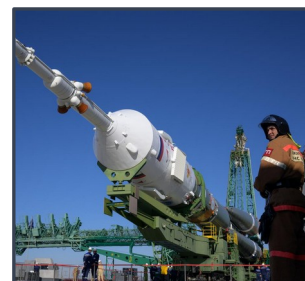
CASE STUDIES



ISS Zvezda, 2019–20

A leak was hunted for over a year and finally found with floating tea leaves.

Source: [universetoday.com](https://www.universetoday.com)



Soyuz MS-22, 2022

A coolant leak from a likely micrometeoroid was seen only once it sprayed.

Source: spacepolicyonline.com



Shenzhou-20, Nov 2025

Debris cracked a window, and it was "unclear when the damage occurred".

Source: theregister.com



NASA impact testing

Piezo sensors located all 147 test impacts within 8 cm (HDU, 2010).

Source: acoustics.org